8.

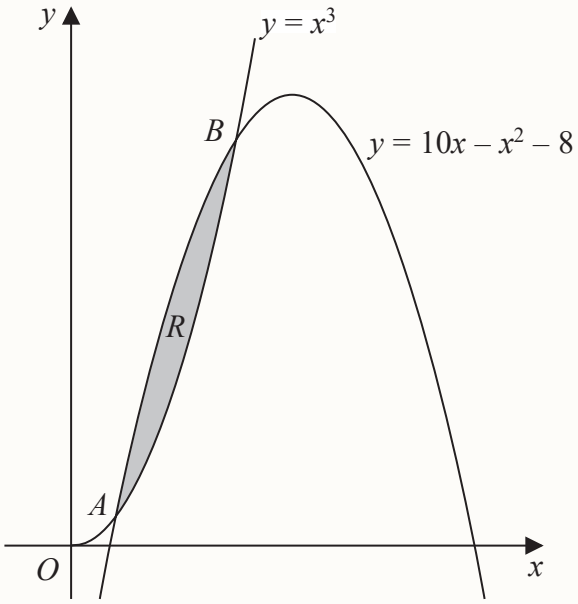


Figure 2

Figure 2 shows a sketch of part of the curves C_1 and C_2 with equations

$$C_1: y = 10x - x^2 - 8 \qquad x > 0$$

$$C_2: y = x^3 \qquad x > 0$$

The curves C_1 and C_2 intersect at the points A and B.

(a) Verify that the point A has coordinates (1, 1)

(1)

(b) Use algebra to find the coordinates of the point B

(6)

The finite region R is bounded by C_1 and C_2

(c) Use calculus to find the exact area of R

(5)